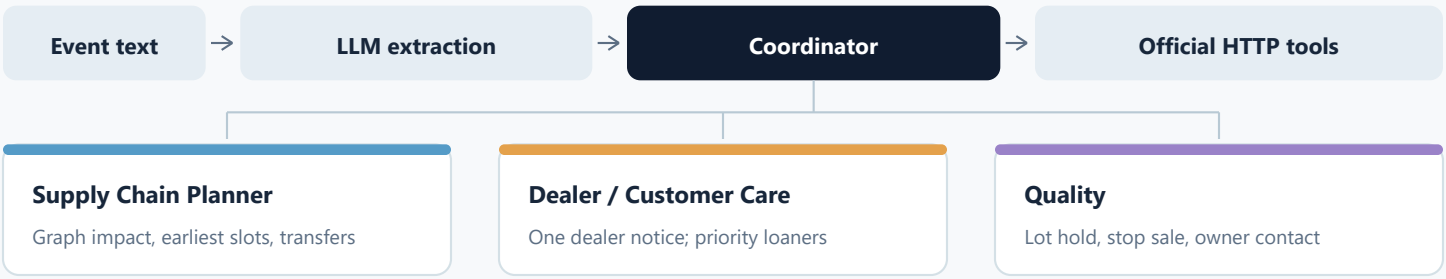
TAMU DATATHON 2026 / CHALLENGE 6 / HITACHI

Highway Control Tower

A traceable response to every disruption.

15/15	0	82/82	814/814
independent event audit	**failed enterprise calls**	**tests passing**	**direct-table graph checks**
Do-nothing baseline: 3/15	900 official HTTP calls	Policy, parser, client, agents	Baseline: no graph

ONE COORDINATOR. THREE BOUNDED SPECIALISTS.



Fresh world per event. Compound incidents share slots, pools and the affected-order union.

LLM UNDERSTANDS. CODE DECIDES.

LLM: classify and split messages; extract entities, confirmation and date phrases. Preserve every raw response.

Code: resolve aliases and relative dates; traverse the BOM and graph; choose affected records, earliest slots and loaner recipients.

Priority: Platinum, then Gold, earlier promised date, lower order ID. Each dealer receives one event-wide notice.

GUARDRAILS BEFORE WRITES

Read live schedules and loaner pools before writes. Validate schema, identity, dates and affected scope.

Never retry an uncertain write. Noise makes no calls; rumors only escalate. Quality separates unsold vehicles from sold owners.

Retain graph paths, decision rationale, tool results and the authoritative action log. No hard-coded event answers.

Company graph | 16 tables | 11 entity types | 16,979 nodes | 28,190 edges

Shipment → Part → Model → ProductionOrder → CustomerOrder → Dealer / Customer

NetworkX graph + actual edge evidence. CSV joins independently check the runtime traversal.

MEASURED EXECUTION AND MODEL PROVENANCE

900 calls: 766 operational writes + 132 free reads + 2 escalations. Final HTTP execution: 10.109s; dry run: 11.064s. These are observed runs, not a comparative benchmark.

Gemini 3.5 Flash-Lite (gemini-3.5-flash-lite): 12,986 input / 1,306 output tokens across 15 responses. The clean run reuses recorded raw output after general normalization; it is not an independent model reproduction.

Google model documentation: https://ai.google.dev/gemini-api/docs/models/gemini-3.5-flash-lite

Limits: local audit is not a hidden judging score. Portal upload remains human-only.

Next: calibrated uncertainty and more paraphrase holdouts. Demo screenshots predate the plants.csv transfer tie-break (4 transfer destinations differ).